Theory of Elementary Particles: Lecture 4

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Charge Conjugation

Time-invariance
- Non-relativistic QM
- Anti-unitary transformation
- $\mathcal{T}$-properties
- Spin
- Dirac Field

Dirac Field and Symmetries

Discrete Symmetries in Interactions
- T-invariance
- Charge Conjugation in Interactions

CPT

Charge Conjugation

$q \leftrightarrow -q$

- Suppose the field in first-quantized picture transforms as

$$\phi \rightarrow e^{-i\alpha q}\phi \tag{1}$$

  – $U(1)$ symmetry, some generalized quantum number which is conserved
- in the second-quantized language:

$$U\phi U^{-1} = e^{-i\alpha q}\phi, \quad U = e^{iQ\alpha} \tag{2}$$

- $$\langle final \mid S \mid in\rangle \approx \left[1 + i\alpha(Q_{in} - Q_{final})\right] \langle final \mid S \mid in\rangle \tag{3}$$

  invariance implies $Q_{in} = Q_{final}$
- the charges of free particles are quantized in units of $|e|$ (confined quarks carry fractional charges)

Charge conjugation in QFT

particle $\leftrightarrow$ antiparticle, can be also realized by unitary operation

$$\mathcal{C}a(\mathbf{p},s)\mathcal{C}^{-1} = b(\mathbf{p},s) \quad (4)$$

$$\mathcal{C}b(\mathbf{p},s)\mathcal{C}^{-1} = a(\mathbf{p},s) \quad (5)$$

$$\mathcal{C}\mathcal{C}^{\dagger} = 1 \quad (6)$$

Useful relation $\rightarrow$ exercise

$$u(\mathbf{p},s) = -i\gamma^2 v^*(\mathbf{p},s) \quad (7)$$

$$v(\mathbf{p},s) = -i\gamma^2 u^*(\mathbf{p},s) \quad (8)$$

$$
\begin{aligned}
* &= \mathcal{C}\psi(t,\mathbf{x})\mathcal{C}^{-1} && (9)\\
&= \mathcal{C}\frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{a(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx}+b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}\mathcal{C}^{-1} && (10)\\
&= \frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{\mathcal{C}a(\mathbf{p},s)\mathcal{C}^{-1}u(\mathbf{p},s)e^{-ipx}+\mathcal{C}b^\dagger(\mathbf{p},s)\mathcal{C}^{-1}v(\mathbf{p},s)e^{ipx}\right\}\\
&= \frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{b(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx}+a^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}\\
&= -i\gamma^2\frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{b(\mathbf{p},s)v^*(\mathbf{p},s)e^{-ipx}+a^\dagger(\mathbf{p},s)u^*(\mathbf{p},s)e^{ipx}\right\}\\
&= -i\gamma^2\frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{a(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx}+b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}^*\\
&= -i\gamma^2\psi^*\\
&= -i\gamma^2(\psi^\dagger)^T\\
&= -i(\overline{\psi}\gamma^0\gamma^2)^T && (11)
\end{aligned}
$$

$$\mathcal{C}\psi(t,\mathbf{x})\mathcal{C}^{-1} = -i(\overline{\psi}\gamma^0\gamma^2)^T \tag{12}$$

- Suppose that similarly as $\mathcal{P}$ and $\mathcal{C}$ $\mathcal{T}$- time-invariance can be realized by unitary transformation U
- Let us consider the Schroedinger equation (for simplicity we write $\psi(t)$ instead of $\psi(t, \mathbf{x})$):

$$i\frac{\partial\psi(t)}{\partial t} = H\psi(t) \tag{13}$$

- $t \to t' = -t$

$$i\frac{\partial\psi(-t)}{\partial t} = H\psi(-t) \tag{14}$$

$$-i\frac{\partial\psi(t')}{\partial t'} = H\psi(t') \tag{15}$$

- It means that $H \to -H$, then the wave function for freely moving particle

$$\phi(t, \mathbf{x}) = \exp(i\mathbf{p} \cdot \mathbf{x} - iEt)$$

goes to

$$\exp(i\mathbf{p} \cdot \mathbf{x} + iEt),$$

hence the positive energy states transform to negative energy states!

$$| E > 0\rangle \to | E < 0\rangle$$

But the states of negative energy are unstable so the assumption that time-invariance is given by $\mathcal{T}\psi(t) = \psi(-t)$ is incorrect!

* Wigner (1932)

- It is assumed that there exists unitary operator U such that

$$H = UH^*U^\dagger \tag{16}$$

- Now take the complex conjugate of the Schroedinger equation

$$-i\frac{\partial \phi^*(t)}{\partial t} = H^*\phi^*(t) \tag{17}$$

- then $t \to t' = -t$

$$i\frac{\partial \phi^*(t')}{\partial t'} = H^*\phi^*(t') \tag{18}$$

  - then

$$i\frac{\partial U\phi^*(t')}{\partial t'} = UH^*\phi^*(t') \tag{19}$$

$$i\frac{\partial U\phi^*(t')}{\partial t'} = UH^*U^\dagger U\phi^*(t') \tag{20}$$

$$i\frac{\partial U\phi^*(t')}{\partial t'} = HU\phi^*(t') \tag{21}$$

- it is seen that $\mathcal{T}$ is realized as i) conjugation and ii) unitary operation

- time-inversion operator on state of vector space is the product of two operators:
  - a unitary transformation U, which replaces the state vector on which it operates with time-reversed state vector;
  - and complex conjugation K of all coefficients that may come with the state-vector

$$\mathcal{T}\alpha\psi(t) = UK\alpha\psi(-t) = \alpha^* U\psi^*(-t), \tag{22}$$

  where α is c-number
  - notice that

$$K^2 = 1, \quad \mathcal{T}^{-1} = KU^\dagger \tag{23}$$

Properties of $\mathcal{T}$

A) antilinearity:

$$\mathcal{T}(a \mid \psi\rangle + b \mid \phi\rangle) = a^* \mathcal{T} \mid \psi\rangle + b^* \mathcal{T} \mid \phi\rangle \tag{24}$$

B) antiunitarity:

$$\langle \mathcal{T}\phi(t) \mid \mathcal{T}\psi(t)\rangle = \langle \phi(-t) \mid \psi(-t)\rangle^*, \tag{25}$$

a distinctive property of $\mathcal{T}$; but the norms of vectors and probabilities remain invariant, just as in the case of unitary operators;

C) operator transformation: $\mathcal{O}' = \mathcal{T}\mathcal{O}\mathcal{T}^{-1} = U\mathcal{O}^*U^\dagger$ for arbitrary operator $\mathcal{O}$, so that

$$\langle \mathcal{T}\phi(t) \mid \mathcal{O}' \mid \mathcal{T}\psi(t)\rangle = \langle \phi(-t) \mid \mathcal{O} \mid \psi(-t)\rangle^* \tag{26}$$

Spin Transformation

By extension of the transformation rule for orbital angular-momentum is assumed that the spin transforms as:

$$\mathcal{T}\mathbf{S}\mathcal{T}^{-1} = -\mathbf{S},$$

where

$$\mathbf{S} = \frac{\boldsymbol{\tau}}{2}$$

$$\tau_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \tau_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \tau_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned} \mathcal{T}\tau_x\mathcal{T}^{-1} &= U(+\tau_x)U^{-1} \to -\tau_x \\ \mathcal{T}\tau_y\mathcal{T}^{-1} &= U(-\tau_y)U^{-1} \to -\tau_y \end{aligned}$$

$$\mathcal{T}\tau_z\mathcal{T}^{-1} = U(+\tau_z)U^{-1} \to -\tau_z$$

- U commutes with τ_y and anti-commutes with τ_x and τ_z so $U = \xi\tau_y$, $UU^\dagger = 1$ hence $|\xi|^2 = 1$. We choose $\xi = -i$.

Now

$$\chi_{\frac{1}{2}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_{-\frac{1}{2}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{27}$$

It is easy to see that

$$-i\tau_y\chi_m = (-1)^{\frac{1}{2}-m}\chi_{-m}, \quad -i\tau_y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{28}$$

then

$$\mathcal{T}|\mathbf{p}, m\rangle = (-1)^{\frac{1}{2}-m}|-\mathbf{p}, -m\rangle \tag{29}$$

Dirac field

Operator transformation

$$\mathcal{T} a(\mathbf{p}, s) \mathcal{T}^{-1} = (-1)^{\frac{1}{2}-s} a(-\mathbf{p}, -s) \tag{30}$$

$$\mathcal{T} b(\mathbf{p}, s) \mathcal{T}^{-1} = (-1)^{\frac{1}{2}-s} b(-\mathbf{p}, -s) \tag{31}$$

Useful relation

$$\gamma^1 \gamma^3 u(\mathbf{p}, s) = -(-1)^{\frac{1}{2}-s} u^*(-\mathbf{p}, -s) \tag{32}$$

$$\gamma^1 \gamma^3 v(\mathbf{p}, s) = -(-1)^{\frac{1}{2}-s} v^*(-\mathbf{p}, -s) \tag{33}$$

$$\mathcal{T}\psi(t,\mathbf{x})\mathcal{T}^{-1} = \frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}}\sum_s \mathcal{T}\left\{a(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx} + b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}\mathcal{T}^{-1} \tag{34}$$

$$.. = \frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}}\sum_s \left\{\mathcal{T}a(\mathbf{p},s)\mathcal{T}^{-1}u^*(\mathbf{p},s)e^{ipx} + \mathcal{T}b^\dagger(\mathbf{p},s)\mathcal{T}^{-1}v^*(\mathbf{p},s)e^{-ipx}\right\}$$

$$= \frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}}\sum_s (-1)^{\frac{1}{2}-s}\left\{a(-\mathbf{p},-s)u^*(\mathbf{p},s)e^{ipx} + b^\dagger(-\mathbf{p},-s)v^*(\mathbf{p},s)e^{-ipx}\right\}$$

$$\mathbf{p} \rightarrow -\mathbf{p} \Rightarrow \tilde{x} = (-t,\mathbf{x}), \quad s \rightarrow -s, \quad (-1)^{\frac{1}{2}-s} \rightarrow (-1)^{\frac{1}{2}+s} = -(-1)^{\frac{1}{2}-s}$$

$$p\cdot x = E_p t - \mathbf{p}\cdot\mathbf{x} \rightarrow E_p t + \mathbf{p}\cdot\mathbf{x} = -p\cdot\tilde{x}$$

$$... = -\frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}}\sum_s (-1)^{\frac{1}{2}-s}\left\{a(\mathbf{p},s)u^*(-\mathbf{p},-s)e^{-ip\tilde{x}} + b^\dagger(\mathbf{p},s)v^*(-\mathbf{p},-s)e^{ip\tilde{x}}\right\}$$

$$= \gamma^1\gamma^3\frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}}\sum_s \left\{a(\mathbf{p},s)u(\mathbf{p},s)e^{-ip\tilde{x}} + b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ip\tilde{x}}\right\} \tag{35}$$

$$= \gamma^1\gamma^3\psi(-t,\mathbf{x}) \tag{36}$$

Summary

$$\mathcal{P}\psi(t,\mathbf{x})\mathcal{P}^{-1} = \eta_P\gamma_0\psi(t,-\mathbf{x}) \quad (37)$$

$$\mathcal{C}\psi(t,\mathbf{x})\mathcal{C}^{-1} = -i\xi_C(\overline{\psi}\gamma^0\gamma^2)^T \quad (38)$$

$$\mathcal{T}\psi(t,\mathbf{x})\mathcal{T}^{-1} = \zeta_T\gamma^1\gamma^3\psi(-t,\mathbf{x}) \quad (39)$$

* The phase factors η_P, ξ_C and ζ_T were omitted in previous derivations.

		P	C	T	CP	CPT
	Γ	$\gamma_0\Gamma\gamma_0$	$\gamma_2\gamma_0\Gamma^T\gamma_2\gamma_0$	$\gamma^1\gamma^3\Gamma^*\gamma^3\gamma^1$	$-\gamma_2\Gamma^T\gamma_2$	$\gamma_5\Gamma\gamma_5$
S	1	1	1	1	1	1
P	$i\gamma_5$	$-i\gamma_5$	$i\gamma_5$	$-i\gamma_5$	$-i\gamma_5$	$i\gamma_5$
V	γ^μ	γ_μ	$-\gamma^\mu$	γ_μ	$-\gamma_\mu$	$-\gamma^\mu$
A	$\gamma^\mu\gamma_5$	-$\gamma_\mu\gamma_5$	$\gamma^\mu\gamma_5$	$\gamma_\mu\gamma_5$	$-\gamma_\mu\gamma_5$	$-\gamma^\mu\gamma_5$
T	$\sigma^{\mu\nu}$	$\sigma_{\mu\nu}$	$-\sigma^{\mu\nu}$	$-\sigma_{\mu\nu}$	$-\sigma_{\mu\nu}$	$\sigma^{\mu\nu}$
∂_μ	∂_μ	∂^μ	∂_μ	$-\partial^\mu$	∂^μ	$-\partial_\mu$

To analyze the symmetry property we consider $\overline{\psi}(x)\Gamma\psi(x)$. The position of the Lorentz index is important, e.g. $\gamma_\mu = (\gamma_0, -\gamma^i)$.

Photon Parity

The photon field reads,

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{r=1}^{2} \left(a(\mathbf{p}, r)\epsilon_\mu(\mathbf{p}, r)e^{-ip\cdot x} + a^\dagger(\mathbf{p}, r)\epsilon^*_\mu(\mathbf{p}, r)e^{ip\cdot x} \right) \Bigg|_{p_0=|\mathbf{p}|} .$$

it can be shown that

$$\mathcal{P}a(\mathbf{p}, \lambda)\mathcal{P}^{-1} = -a(-\mathbf{p}, -\lambda) \quad \text{(helicity basis: helicity flips under parity)}$$

so the photon has negative parity!

QED is Parity Invariant

$$\mathcal{L}_{QED}(x) = \overline{\psi}(x)\left(i\gamma^\mu\partial_\mu - M\right)\psi(x) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \underbrace{-e\overline{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)}_{\mathcal{L}_{int}(x)=-\mathcal{H}_{int}(x)} \qquad (40)$$

We keep the normalization $e = -|e|$

- Notice that

$$\mathcal{P}A^\mu(x^\mu)\mathcal{P}^{-1} = A_\mu(x_\mu),$$

and

$$\begin{aligned}\mathcal{P}\overline{\psi}(x)\mathcal{P}^{-1} &= \mathcal{P}\,\psi^\dagger(x)\mathcal{P}^{-1}\gamma^0 = (\mathcal{P}\,\psi(x)\mathcal{P}^{-1})^\dagger\gamma^0 = (\gamma^0\psi(x'))^\dagger\gamma^0 = \psi^\dagger(x')\gamma^0\gamma^0 \\ &= \overline{\psi}(x')\gamma^0,\end{aligned}$$

where $x'^\mu = x_\mu$ and $\mathcal{P}^\dagger = \mathcal{P} = \mathcal{P}^{-1}$

- The free-field part:

$$\begin{aligned}\mathcal{P}\overline{\psi}(x)\gamma^\mu\partial_\mu\psi(x)\mathcal{P}^{-1} &= \mathcal{P}\overline{\psi}(x)\mathcal{P}^{-1}\gamma^\mu\partial_\mu\mathcal{P}\psi(x)\mathcal{P}^{-1} = \overline{\psi}(x')\gamma^0\gamma^\mu\gamma^0\partial_\mu\psi(x') \\ &= \overline{\psi}(x')\gamma_\mu\partial'^\mu\psi(x')\end{aligned}$$

where $\gamma^0\gamma^\mu\gamma^0 = \gamma_\mu$ and $\partial'_\mu = \partial^\mu$

Parity and Interactions

- $$\mathcal{P}\overline{\psi}(x)\psi(x)\mathcal{P}^{-1} = \mathcal{P}\overline{\psi}(x)\mathcal{P}^{-1}\mathcal{P}\psi(x)\mathcal{P}^{-1} = \overline{\psi}(x')\gamma^0\gamma^0\psi(x') = \overline{\psi}(x')\psi(x')$$
- $$\mathcal{P}\overline{\psi}(x)\gamma^\mu\psi(x)\mathcal{P}^{-1} = \overline{\psi}(x')\gamma_\mu\psi(x')$$
- the interacting part

$$\begin{aligned}\mathcal{P}A_\mu(x)\overline{\psi}(x)\gamma^\mu\psi(x)\mathcal{P}^{-1} &= \mathcal{P}A_\mu(x)\mathcal{P}^{-1}\mathcal{P}\overline{\psi}(x)\mathcal{P}^{-1}\gamma^\mu\mathcal{P}\psi(x)\mathcal{P}^{-1} \\ &= A_\mu(x')\overline{\psi}(x')\gamma^\mu\psi(x')\end{aligned}$$

- Maxwell part

$$\begin{aligned}\mathcal{P}F_{\mu\nu}(x)\mathcal{P}^{-1} &= \mathcal{P}(\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x))\mathcal{P}^{-1} \\ &= \partial_\mu\mathcal{P}A_\nu(x)\mathcal{P}^{-1} - \partial_\nu\mathcal{P}A_\mu(x)\mathcal{P}^{-1} \\ &= \partial'^\mu A^\nu(x') - \partial'^\nu A^\mu(x')\end{aligned}$$

hence

$$\mathcal{P}F_{\mu\nu}(x)F^{\mu\nu}(x)\mathcal{P}^{-1} = \mathcal{P}F_{\mu\nu}(x)\mathcal{P}^{-1}\,\mathcal{P}F^{\mu\nu}(x)\mathcal{P}^{-1} = F_{\mu\nu}(x')F^{\mu\nu}(x')$$

Parity and Interactions

$$\mathcal{P}\mathcal{L}_{QED}(x)\mathcal{P}^{-1} = \mathcal{L}_{QED}(x')$$

Notice that

$$\mathcal{P}: \quad S = \int d^4x \mathcal{L}_{QED}(x) \to \int d^4x \mathcal{L}_{QED}(x') = \int d^4x' \mathcal{L}_{QED}(x')$$

- QED is invariant under $\mathcal{P}$!But it is the result of construction ?!
- Using the same arguments one can show that QCD lagrangian is invariant under $\mathcal{P}$: strong interactions are also $\mathcal{P}$-invariant
- Weak interactions break $\mathcal{P}$ symmetry!

Intrinsic Parity

- One assumes that the ground state is the positive parity state. The ground state is the reference point for other consideration.
- For the particle having non-vanishing additive quantum numbers, it is necessary first to fix the parities of reference particles

$$\eta_{neutron} = \eta_{proton} = \eta_{\Lambda} = 1$$

π^- parity – pion capture by deuteron: W. Chinowsky, J. Steinberger, *Absorption of Negative Pions in Deuterium: Parity of the Pion*. Phys.Rev. 95 (1954) 1561
We shall establish the π^- parity

- consider pion capture by deuteron

$$\pi^- + d \to n + n$$

Deuteron parity

- Deuteron is typically in ${}^3S_1$ state, ${}^{2s+1}l_J$, s-spin, l-angular momentum, J total angular momentum, so $s = 1$, $J = 1$, $l = 0$ (S-wave)
* partial waves: S, P, D, correspond to $l = 0, 1, 2$ and parities $+1, -1, +1$
- Note that $\eta_n = \eta_p = 1$, hence the deuteron parity reads

$$\eta_d = \eta_n \eta_p (-1)^{l=0} = 1 \tag{41}$$

- From parity conservation (initial system parity = final system parity)

$$\eta_\pi \eta_d (-1)^{l_{\pi d}} = \eta_n \eta_n (-1)^{l'_{nn}} \tag{42}$$

  where $l_{\pi d}$ and l'_{nn} are relative angular moments of the particles initial and final respectively

- π^- is slowed down and captured so the system is in $l_{\pi d} = 0$ S-state (ground), hence

$$\eta_\pi = (-1)^{l'_{nn}} \tag{43}$$

- Initial states: $s_d = 1$ **(deuteron spin),** $s_\pi = 0$ **(pion spin), relative** $\mathbf{L} = 0 \Rightarrow$ **the total angular momentum of the deuteron-pion system reads** $J = 1$, now, we need to find the angular momentum of neutron-neutron system !

- The total angular momentum is conserved!
- Let us look at final states:
- if the relative angular momentum of system of two neutrons is equal to $\mathbf{L}$ and the total spin reads $\mathbf{s}$ then the total angular momentum reads

$$\mathbf{J} = \mathbf{s} + \mathbf{L} \tag{44}$$

Symmetry of neutron-neutron system

- nonrelativistic wave function

$$\psi = \phi(space) \times \alpha(S, S_z)$$

- symmetric spin-1 case

$$\alpha(1,1) = \uparrow\uparrow \tag{45}$$

$$\alpha(1,0) = \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow) \tag{46}$$

$$\alpha(1,-1) = \downarrow\downarrow \tag{47}$$

  interchange between neutrons gives $(-1)^{1+1} = 1$

- anti-symmetric spin-0 case

$$\alpha(0,0) = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow) \tag{48}$$

  interchange between neutrons gives $(-1)^{1+0} = -1$

Symmetry of neutron-neutron system

- The total angular momentum of the πd must be $J = 1$
- Relative angular momentum (neutron-neutron) and the spin:
  - # J could be $|L - s|$,...,$|L + s|$
  - $L = 0$, $s = 1$ (symmetric)
  - $L = 1$, $s = 0$ (antisymmetric)
  - $L = 1$, $s = 1$ (symmetric)
  - $L = 2$, $s = 1$ (symmetric)
  - * it generates the symmetry factor $(-1)^{1+s}$ due to neutron – neutron interchange
- neutrons are fermions so total wave function should be totally antisymmetric when neutron $\rightleftarrows$ neutron, from $\mathbf{x} \rightleftarrows -\mathbf{x}$ (in cms of neutron neutron) $(-1)^l$. The total result of interchange between neutron-neutron reads

$$(-1)^{1+s+L} = -1 \tag{49}$$

  As the result $s + L$ must be even!
- the only even combination is
  $L = 1$, $s = 1$ so it is ${}^3P_1$ state and its parity is $(-1)^1 = -1$
- **Conclusion the parity of pion reads:**

$$\eta_\pi = (-\mathbf{1})^{\mathbf{1}} = -\mathbf{1} \tag{50}$$

π^0 Parity

- consider

$$\pi^0 \to \gamma(\mathbf{k}, \boldsymbol{\epsilon}_1) + \gamma(-\mathbf{k}, \boldsymbol{\epsilon}_2) \tag{51}$$

  decay, where $\mathbf{k}$ denotes momentum, while $\boldsymbol{\epsilon}_{1,2}$ polarization of the $1, 2$ photon
- notice that photons propagating in vacuum obey $\mathbf{k} \cdot \boldsymbol{\epsilon}_{1,2} = 0$
- initial π^0 (at the rest frame) state has $J = 0$, hence, the final state should also have $J = 0$

- The final state is a product of two bosonic states, so the total wave function must be symmetric.
- An exchange: $\mathbf{k} \leftrightarrow -\mathbf{k}$ and $\boldsymbol{\epsilon}_1 \leftrightarrow \boldsymbol{\epsilon}_2$ must lead to symmetric wave function
- The most general form of the wave function of two photons reads

$$\psi_{2\gamma} = \psi_{2\gamma,odd} + \psi_{2\gamma,even}, \tag{52}$$

where

$$\psi_{2\gamma,even} = A(\boldsymbol{\epsilon}_1 \cdot \boldsymbol{\epsilon}_2) \sim \cos(\phi), \quad \psi_{2\gamma,odd} = B(\boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2) \cdot \mathbf{k} \sim \sin(\phi) \tag{53}$$

ϕ is relative angle between photons polarizations

* The wave function splits into even (η_+) and odd (η_-) contributions with respect to the parity. Indeed, $\cos(\phi)$ and $\sin(\phi)$ are even and odd functions, respectively.

- The interchange between photons does not change the amplitude.

$$\eta_\pi = \eta_\pm \eta_\gamma^2 = \eta_\pm \tag{54}$$

- Note that only **odd-sine** distribution is experimentally observed (perpendicular polarizations!): π^0 is pseudo-scalar -1 parity
- With respect to Parity

$$\eta_+(=+1)A(\boldsymbol{\epsilon}_1 \cdot \boldsymbol{\epsilon}_2) + \eta_-(=-1)B(\boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2) \cdot \mathbf{k} \sim \sin(\phi) \tag{55}$$

Pion parity

In practice the parity of pion was investigated in

$$\pi^0 \to (e^- + e^+) + (e^- + e^+)$$

See: R. Plano, A. Prodell, N. Samios, M. Schwartz, and J. Steinberger, *Parity of the Neutral Pion*, Phys. Rev. Lett. 3, 525 (1959)

T-invariance in interactions

- QED Lagrangian, as defined previously, is invariant to time inversion, however, the electric dipole interaction is not

$$H_d = -\mu^{(el)} \boldsymbol{\sigma} \cdot \mathbf{E}, \tag{56}$$

  ($\mathbf{E} \to \mathbf{E}$ but $\boldsymbol{\sigma} \to -\boldsymbol{\sigma}$)

- $\mu^{(el)}$ must be real if $H_d = H_d^\dagger$
- Experiments have proved that this interaction is negligible, as shown by the extremely small values of the measured electric dipole moment of the typical fermions:

$$e \qquad |d_e| < 4.1 \times 10^{-30}\, e\,\mathrm{cm} \quad (\mathrm{JILA},\ 2023) \tag{57}$$

$$\mu \qquad |d_\mu| < 1.8 \times 10^{-19}\, e\,\mathrm{cm} \quad (\mathrm{BNL}) \tag{58}$$

$$p \qquad |d_p| < 2.1 \times 10^{-25}\, e\,\mathrm{cm} \quad (\mathrm{indirect},\ {}^{199}\mathrm{Hg}) \tag{59}$$

$$n \qquad |d_n| < 1.8 \times 10^{-26}\, e\,\mathrm{cm} \quad (\mathrm{nEDM},\ 2020) \tag{60}$$

- β-decay is time-reversal-invariant but not all weak processes have this property!

- if $\mathcal{T}$-reversal symmetry is violated then CP symmetry must be also violated!
- In the Standard Model $\mathcal{CP}$ is naturally violated if the number of families is larger than **two**: Kobayashi–Maskawa, with three families the quark mixing matrix contains one physical complex phase!
- According to quantum chromodynamics **there can be a violation of CP symmetry in the strong interactions**. However, **there is no experimental evidence for that**. As there is no known reason for it to be conserved in QCD specifically, this is a "fine tuning" problem known as **the strong CP problem**.

Charge Conjugation in Interactions

- Only neutral particles, states, like π^0, η^0, e^+e^-, $p\overline{p}$ and $\pi^+\pi^-$ are eigenstates of $\mathcal{C}$, indeed

$$\mathcal{C}|\pi^+\rangle = |\pi^-\rangle \tag{61}$$

but

$$\mathcal{C}|\pi^0\rangle = C_\pi|\pi^0\rangle \quad C_\pi = \pm 1 \tag{62}$$

- Assuming that electromagnetic interaction is C-invariant the photon is odd $C_\gamma = -1$, indeed:
- Consider a charged fermion field

$$j^\mu = -e\bar{\psi}\gamma^\mu\psi \tag{63}$$

- Transformation of Noether current associated with global gauge transformations $U(1)$, due to $\mathcal{C}$, reads

$$\begin{aligned} \mathcal{C}j^\mu\mathcal{C}^{-1} &= -e\mathcal{C}\bar{\psi}\gamma^\mu\psi\mathcal{C}^{-1} = +e\,\bar{\psi}\gamma^\mu\psi & (64)\\ &= -j^\mu & (65)\end{aligned}$$

  see the table of bilinear transformations (section: Dirac Field and Symmetries)
- if $j_\mu A^\mu$ is $\mathcal{C}$-invariant then

$$\mathcal{C}A_\mu\mathcal{C}^{-1} = -A_\mu \tag{66}$$

- n-photon system has charge parity $(-1)^n$

- Furry's theorem: *the matrix element of an operator invariant to charge conjugation vanishes if the number of external photons is odd and if there are no other particles!*

$$
\begin{aligned}
* &= \langle f|\mathcal{C}^{-1}\mathcal{C}\mathcal{H}_{int}\mathcal{C}^{-1}\mathcal{C}|i\rangle && (67)\\
&= (-1)^{n_f+n_i}\langle f|\mathcal{H}_{int}|i\rangle && (68)
\end{aligned}
$$

$n_{f,i}$ number of photons in the final/initial state

-

$$
\begin{aligned}
\pi^0 &\rightarrow 2\gamma\ (\text{branching ratio}:\ 98.8\%)\\
\eta^0 &\rightarrow 2\gamma\ (\text{branching ratio}:\ 39.4\%)
\end{aligned}
$$

$C_\pi = C_\eta = 1$

$$
\begin{aligned}
\frac{\pi^0 \rightarrow 3\gamma}{\pi^0 \rightarrow 2\gamma} &< 3\cdot 10^{-8}\\
\frac{\eta^0 \rightarrow 3\gamma}{\eta^0 \rightarrow 2\gamma} &< 5\cdot 10^{-4}
\end{aligned}
$$

Parities

We say that π and η are $I(J^{PC}) = 1(0^{-+})$ and $0(0^{-+})$ particles, i.e. pseudoscalars with charge parity $+1$.

CPT theorem, by Lüders, Pauli and Schwinger

CPT theorem

The product of the transformations $\mathcal{C}$, $\mathcal{P}$, $\mathcal{T}$ applied in any order is always a symmetry of a quantum theory if the Lagrangian that defines it is Hermitian, invariant under Lorentz transformations, is built up from normal-ordered products of fields, and if the fields are quantized in accord with the usual spin-statistic connection.

$$\theta\mathcal{L}(x)\theta^{-1} = \mathcal{L}^{\dagger}(-x) \to \mathcal{L}(-x), \quad \theta = \mathcal{CPT} \tag{69}$$

for Hermitian Lagrangian.
Notice that action remains unchanged

$$\int d^4x\mathcal{L}(-x) = \int d^4x\mathcal{L}(x) \tag{70}$$

The validity of the theorem is based on the invariance to the group of continuous Lorentz transformations, the usual spin-statistics connection and the locality of the theory.

Let $\theta = \mathcal{CPT}$, then:

- all complex numbers are replaced by their complex conjugates;
- space-time coordinates x^μ are replaced by $x'^\mu = -x^\mu$;
- scalar fields are transformed as $\theta\phi(x)\theta^{-1} = \omega_B\phi(-x)$, where ω_B is fixed to be $+1$;
- $\theta A_\mu(x)\theta^{-1} = -A_\mu(-x)$
- $\theta\psi(x)\theta^{-1} = \omega_F\gamma_5\gamma_0\overline{\psi}^T(-x)$, $|\omega_F|^2 = 1$
- bilinear products of fermion fields $\overline{\psi}\Gamma\psi$ may be grouped according to their transformations under θ: $\Gamma_+ = \{1, \sigma^{\mu\nu}, i\gamma_5\}$ are even (do not change a signs), and $\Gamma_- = \{\gamma^\mu, \gamma^\mu\gamma_5\}$ are odd (change signs)

Implications of CPT

$$\theta|a\rangle = e^{i\varphi}|\overline{a}\rangle \tag{71}$$

in the invariant theory

$$\langle a|H|a\rangle = \langle \overline{a}|H|\overline{a}\rangle \tag{72}$$

the energy spectra in the original and the transformed systems are identical. In practice in the absence of interactions $\langle a|H|a\rangle$ gives the masses of particles hence $\mathcal{CPT}$-invariance implies the equality between masses of the particles and corresponding anti-particles:

$$\frac{m_p - m_{\overline{p}}}{m_p} \quad \approx \quad 2 \times 10^{-11}. \tag{73}$$